

Introduction to Engineering Technology

A Comprehensive Study on Technological Evolution, Management, and Industrial Applications

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1. Definition of Engineering Technology

Engineering Technology is defined as the discipline and process of applying engineering principles to formulate and deliver practical solutions for real-world problems (existing or potential) to enhance human well-being and societal development.

2. Evolution of Technology

Technology's evolution is categorized based on technological utilization, cultural transformations, and landmark innovations across different eras in human civilization.

Primary Technological Eras

- 1. Stone Age
- 2. Pastoral / Herding Era
- 3. Agricultural Era
- 4. Industrial Era
- 5. Information Technology Era

Detailed Timeline of Historical Eras and Innovations

Era	Time Period	Key Technological Innovations & Uses
Stone Age	Up to c. 6000 BCE or 4000 BCE	Discovery and control of fire, stone tools, use of skin/leather bags for goods transportation.
Bronze Age	c. 4000 BCE – c. 1200 CE	Development of the wheel, agricultural implements, animal-driven transport, carts, bows and metal-tipped arrows.
Iron Age	c. 1200 BCE – c. 500 CE	Arch-structured architecture, steel weapons, metal tools, iron-rimmed wheels/horse tack, saddles, cannons, sailing and paddled ships.

Era	Time Period	Key Technological Innovations & Uses
Middle Ages	c. 500 CE – c. 1450 CE	Advancements in metal weaponry, specialized saddlery, artillery, and oceanic navigation vessels.
Renaissance / Colonial Era	c. 1450 CE – c. 1700 CE	Luxury ocean liners, firearms/guns, hot air balloons, specialized warships.
Industrial Revolution	c. 1700 CE – c. 1900 CE	Steam engines, railway transport systems, direct electric current (DC).
World Wars Era	c. 1900 CE – c. 1950 CE	Automobiles, aircraft, submarines, nuclear bombs, fighter aircraft, radio communication devices.
Modern Era	c. 1950 CE – c. 2000 CE	Electronic computers, rocketry, spacecraft, scanning technology, solar panels, space shuttles.
Space Exploration Era	c. 2000 CE – Present	Cloning, nanotechnology, advanced space exploration.

3. The Industrial Revolution & Major Turning Points

Key Characteristics of the Industrial Revolution

- Transition from manual labor and handcrafts to machinery.
- Harnessing water and steam power for industrial production.
- Initiation of coal usage as a primary fuel source.
- Manufacturing of steam-powered ships and locomotives.
- Emergence of large-scale mass production and dedicated machinery.

Key Turning Points in Technology History

- Control of Fire
- Invention of the Wheel
- Water Wheel
- Cast Iron Casting
- Printing Press
- Static Electricity
- Electrical Conduction
- Telephone
- Filament Lamp (Light Bulb)
- Wireless Communication
- Electronic Valves (Vacuum Tubes)
- Transistors & Integrated Circuits (ICs)
- Development of Cement & Glass

4. Impact of Engineering Technology

I. Impact on Humans and Society

- **Demographic Distribution:** Shifts in population density and urbanization patterns influenced by technological hubs.
- **Capital Ownership:** Changes in economic structures and capital accumulation.
- **Productivity:** Significant increases in manufacturing capacity, throughput, and efficiency.
- **Healthcare & Security:** Advanced medical diagnostic systems and enhanced safety/defense systems.
- **Social & Cultural Constructs:** Transformation of family relationships, information access, and cultural habits.

II. Environmental Impact

- Environmental pollution (air, water, soil).
- Global warming and the greenhouse effect.
- Depletion of natural finite resources.
- Creation of new artificial/built environments.

5. Future Trends & Product Enhancements

Future Trends in Technology

- Transition towards renewable and sustainable energy resources.
- Development and adoption of eco-friendly, low-pollution devices.
- Expanded reliance on wireless communication protocols.
- Advanced product optimization, commercialization, and technology management.

Criteria for Evaluating Product Improvements

When modernizing products, structural and functional upgrades are measured against key standards:

Operational Quality

Accurate functionality, high efficiency, multi-functionality, interchangeability, and high-quality finishing.

Safety & Compliance

Ergonomics/human factors engineering, health & safety standards, eco-friendliness, and quality standard compliance.

Drivers of Product Improvement

1. Need for competitiveness in market dynamics.
2. Personal and consumer demands.
3. Government initiatives and policy objectives.

6. Commercialization and Management of Technology

Commercialization

Commercialization refers to fulfilling human needs and wants within a market economy through supply and demand forces, driven by profit motives.

Factors Influencing Market Demand:

- Purchasing power of consumers.
- Total consumer volume.
- Product quality, durability, and modern relevance.
- Human attitudes and cultural traits.
- Environmental context and state policies/regulations.

Management of Technology

Technology management integrates science, technology, and administrative management to plan, develop, and execute technological capabilities to achieve an organization's strategic and operational goals.

1. Planning

Defining objectives and determining strategic action plans to accomplish them.

2. Organizing

Selecting and structuring resources, tools, techniques, and personnel.

3. Designing

Constructing design drawings, schematics, and models to demonstrate product goals.

4. Operation & Control

Engaging personnel in technical processes and monitoring execution to correct deviations.

Key Considerations in Technology Management

- The strategic role of technology within the organization.
- Forecasting technological advancements and trends.
- Mapping technological trajectories.
- Managing project scale and scope.

7. Classification of Local Industries

Industries can be categorized based on various criteria:

By Nature of Production

- **Mining & Quarrying:** Graphite, gemstones, mineral extraction.
- **Material Processing:** Processing of primary agricultural goods (Tea, Rubber, Coconut).
- **Finished Goods Manufacturing:** Ceramics, footwear, wooden furniture.
- **Repair & Maintenance Services:** Automobiles, building maintenance, transport services.

By Scale of Industry

- Small and Medium Scale Enterprises (SMEs)
- Large Scale Enterprises

Important Classification Metrics in Sri Lanka:

- **Industrial Development Board (IDB):** Classifies industry scale primarily based on the amount of *Capital Invested*.
- **Central Bank of Sri Lanka (CBSL):** Classifies industry scale based on the volume of *Electricity Consumption*.

By Location & Mode of Production

Location-Based

- Cottage / Home Industries
- Factory-Based Industries

Mode of Production

- Fully Manual Labor
- Semi-Automated
- Fully Automated